

Autosomal dominant polycystic kidney disease (ADPKD) is one of the most common heterogeneous disorders, with a reported incidence of 1 in 1,000 people. This disorder causes the development of numerous fluid-filled renal cysts that ultimately expand and severely impair kidney function. ADPKD is often considered to be late-onset, with symptoms becoming more prominent in older adults (ie, age 40 and above). By age 60, approximately 50% of patients with ADPKD progress to end-stage renal disease (ESRD) and kidney failure. Clinical complications may include high blood pressure, urinary tract infections, and cysts in other organs such as the liver and pancreas.

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To identify individuals with mutations in *PKD1* and *PKD2*, researchers analyzed phenotypic and genetic data from patients with established renal disease, as shown in Table 1.

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Researchers next examined how phenotypic disease severity was affected by both the position of the mutations in *PKD1* (5' end or 3' end) and the type of mutation in *PKD1* or *PKD2*. In Figure 1, disease severity is given as the cumulative probability of renal survival (ie, probability of maintaining functional kidneys) as a function of age.

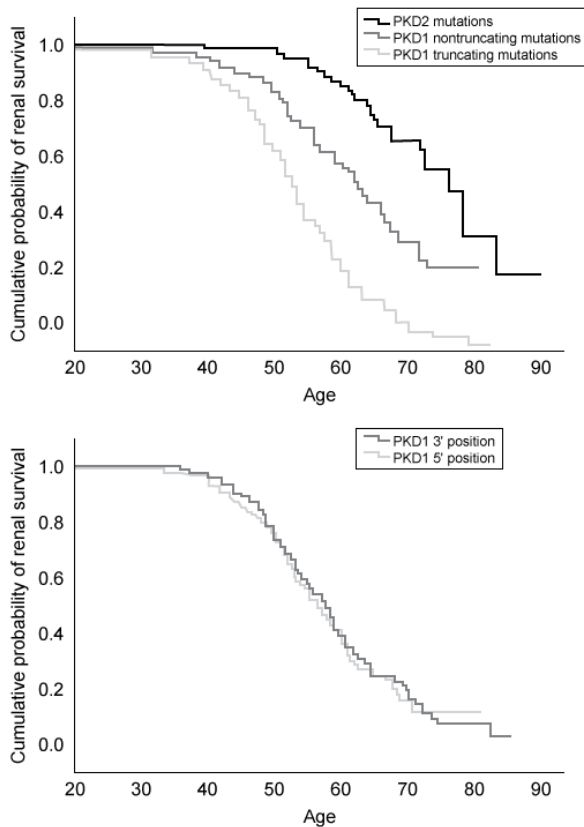


Figure 1 Probability of maintaining functional kidneys during aging, analyzed by mutation type and position

Which of the following DNA mutations would always lead to truncated PC1 if observed in the portion of the *PKD1* coding strand shown below? (Note: 5'-TGG-3' is the first codon in the reading frame shown.)

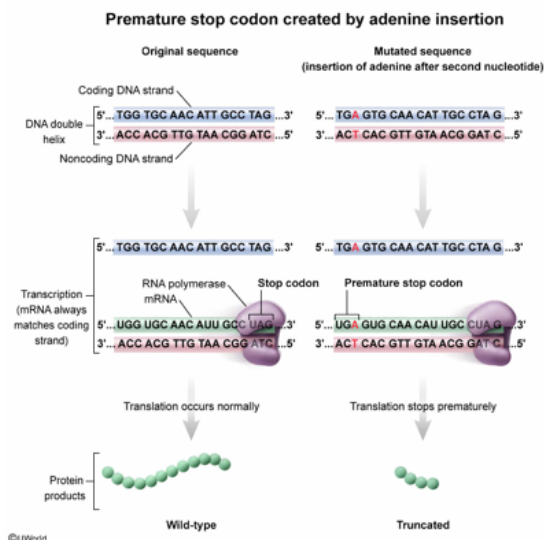
5' — TGGTGCAACATTGCCTAG — 3'

- ☐ A. Substitution of the second nucleotide for a thymine nucleotide [8%]
- ☐ B. Deletion of the guanine nucleotide in the fifth position [43%]
- ☐ C. Addition of three cytosine nucleotides after the third nucleotide [8%]
- ✓ ☒ D. Insertion of an adenine nucleotide after the second nucleotide [38%]

Correct

37% answered correctly

Explanation:



When DNA is **transcribed**, the noncoding DNA strand is read by RNA polymerase to produce a complementary 5' to 3' mRNA strand. As a result, the **mRNA transcript** is always **identical** to the **coding DNA strand** with one exception: **RNA** has uracil instead of thymine. Transcription is followed by **translation** (protein synthesis), a process in which the ribosome reads the mRNA transcript in a 5' to 3' direction until a **stop codon** (UAA, UAG, or UGA) is recognized, causing **translation termination** and polypeptide release.

Mutations are changes in the DNA sequence that arise from errors in replication or from mutagen exposure (man-made or environmental). A **nonsense mutation** is a mutation in which nucleotide change creates a **premature stop codon**, causing early translation termination and production of a truncated (shortened) protein.

In the question, a portion of the *PDK1* gene coding sequence is shown, with the first codon in the **reading frame** being **TGG**. For the sequence shown, the **insertion of an adenine nucleotide** after the **second nucleotide** in the portion of the *PDK1* coding sequence shown would convert this first codon to **TGA**, which corresponds to UGA (ie, a stop codon) in mRNA. This is a *nonsense mutation* that results in a *truncated* PC1.

(Choice A) G→T **substitution** at the second nucleotide does *not* produce a premature stop codon. Instead, this is a missense mutation, meaning the nucleotide change corresponds to a different amino acid.

(Choice B) **Frameshift mutations** result from the insertion or deletion of a number of nucleotides not divisible by 3. These mutations change the reading frame of the transcript, giving rise to proteins with altered amino acid sequences. They may also generate truncated proteins *if* a *premature* stop codon is present in the new reading frame. Deletion of the guanine nucleotide at the **fifth position** changes the reading frame but does *not* generate a premature stop codon.

(Choice C) Insertion of three cytosine nucleotides after the third nucleotide generates neither a nonsense mutation nor a frameshift mutation with a downstream stop codon. Instead, the resulting polypeptide will simply contain one additional amino acid that corresponds to the CCC triplet.

Educational objective:

Various types of DNA mutations can alter the protein product of a gene. Truncated proteins result from nonsense mutations and frameshift mutations with a downstream, premature stop codon (UAA, UAG, or UGA) in the new reading frame.

Time spent:0 sec

Subject:
Biology

QID:400844

System:
Genetics and Evolution

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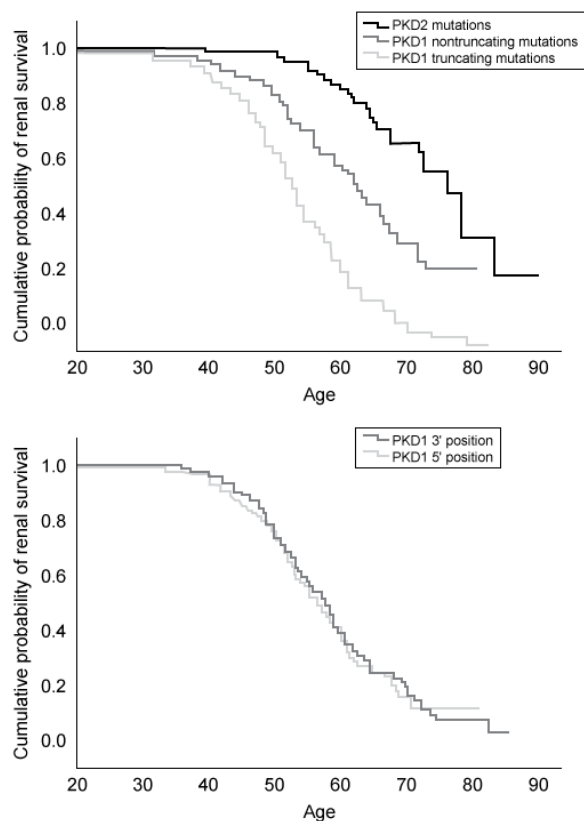


Figure 1 Probability of maintaining functional kidneys during aging, analyzed by mutation type and position

A woman whose father is heterozygous for an ADPKD-causing mutation in *PKD1* only and whose mother is unaffected by ADPKD-causing mutations has a child with a man who has no family history of ADPKD. What is the percent chance that this child will have ADPKD? (Note: Assume this woman is a young adult with unknown ADPKD status.)

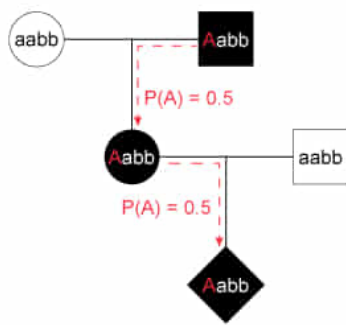
- ☐ A. 0% [19%]
- ☒ B. 25% [56%]
- ☐ C. 50% [23%]
- ☐ D. 100%

Correct

55% answered correctly

Explanation:

Autosomal dominant inheritance of ADPKD



$$\begin{aligned} P(\text{child with ADPKD}) &= P(\text{mother inherited A allele from father}) \\ &\times P(\text{mother passes on the A allele to child}) \\ &= 0.5 \times 0.5 = 0.25 \text{ or } 25\% \end{aligned}$$

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Based on the passage, ADPKD is an **autosomal dominant disease**, signifying that only one copy of a dominant allele is necessary to produce the phenotype. The disease-causing genes, *PDK1* and *PDK2*, are located on **autosomal (nonsex) chromosomes** and, as a result, each parent transmits one of two gene copies to their child. In the given scenario, let the following be true:

- For *PKD1*: *A* is the dominant mutated allele, and *a* is the recessive wild-type allele
- For *PKD2*: *B* is the dominant mutated allele, and *b* is the recessive wild-type allele

The woman's mother is unaffected by ADPKD-causing mutations, and therefore would have the genotype *aabb*. However, the woman's father is heterozygous for a dominant mutation in the *PKD1* gene only (ie, genotype *Aabb*). A **heterozygous parent** has a 50% chance of transmitting the mutation to their offspring. Therefore, for *PKD1*, the father (*Aa*) will either transmit the mutant *A* allele with a 0.5 probability or the wild-type *a* allele with a 0.5 probability. In contrast, the mother (*aa*) can only pass on the wild-type *a* allele to her children. For *PKD2*, each parent contributes a wild-type *b* allele, giving this woman the genotype *bb*. Therefore, this woman's full genotype is either ***aabb*** (normal) or ***Aabb*** (ADPKD).

For this woman to have an ADPKD-affected child with a man who has no family history of ADPKD (ie, genotype *aabb*), two independent events must occur. Specifically, the woman would have to inherit the *A* allele from her father (50% chance) *and* pass this *A* allele onto her child (50% chance). Accordingly, given this woman's current lack of symptoms (ie, unknown genotype), the percent chance of these events happening can be calculated using the probability rule for **independent events**:

$$P(\text{child with ADPKD}) = P(\text{woman inherited A allele from her father}) \times P(\text{woman passes A allele to her child})$$

$$P(\text{child with ADPKD}) = (0.5)(0.5) = 0.25, \text{ or } 25\%$$

(Choices A, C, and D) These options result from *incorrect* calculations.

Educational objective:

In autosomal dominant inheritance, transmission of only one copy of a dominant allele is necessary to produce the phenotype. A heterozygous parent has a 50% chance of transmitting the mutation to their offspring.

Time spent:0 sec

Subject:
Biology

QID:400845

System:
Genetics and Evolution

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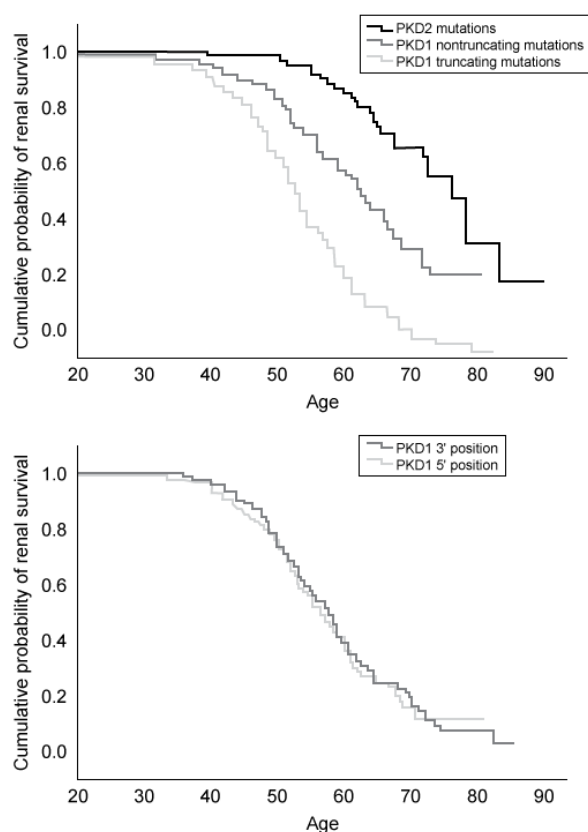


Figure 1 Probability of maintaining functional kidneys during aging, analyzed by mutation type and position

Based on the results presented in Figure 1, which of the following statements about patients with ADPKD are true?

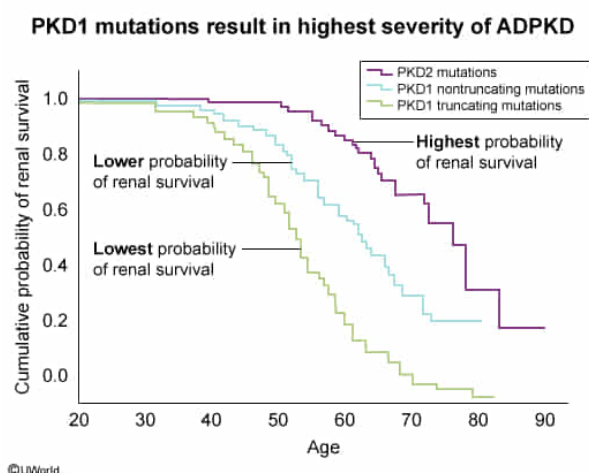
- I. Truncating mutations result in more severe ADPKD symptoms than nontruncating mutations in *PKD1*.
- II. The closer the mutation is to the 5' end of the *PKD2* gene, the more severe the disease phenotype.
- III. *PKD1* mutations result in more severe disease pathology than *PKD2* mutations.

- ☐ A. I only [5%]
- ☐ B. I and II only [2%]
- ☒ C. I and III only [88%]
- ☐ D. I, II, and III [3%]

Correct

87% answered correctly

Explanation:



The two graphs in Figure 1 demonstrate renal survival curves for patients with ADPKD. In both graphs, the x-axis represents age and the y-axis represents the probability of renal survival (ie, the probability that the kidneys remain functional). A decreased probability of renal survival indicates increased ADPKD severity.

The passage states that ADPKD is a late-onset disorder. Therefore, as illustrated in the renal survival curves, the kidneys have a nearly 100% chance of remaining functional from birth through age 40; however, the probability of renal survival begins to decline after age 40. Accordingly, the top graph in Figure 1 shows the following:

- Out of the individuals with *PKD1* mutations, those with truncating mutations have more severe symptoms and are more likely to experience kidney failure (ie, have lower probability of renal survival) after age 40 compared to individuals with nontruncating *PKD1* mutations (**Number I**).
- Individuals with either a truncating or nontruncating *PKD1* mutation have a lower probability of renal survival compared to individuals with *PKD2* mutations. Therefore, ***PKD1* mutations** cause a **more severe disease** phenotype than *PKD2* mutations (**Number III**).

(Number II) The bottom graph of Figure 1 shows *little to no* difference in the probability of kidney failure regardless of whether the mutation is located on the 3' end or 5' end of the *PKD1* gene. As a result, the renal survival curve indicates that disease severity is likely independent of the **location** of the *PKD1* mutation. In addition, Figure 1 provides *no* information regarding the locations of *PKD2* mutations.

Educational objective:

A survival curve illustrates the proportion of healthy individuals remaining in a population (y-axis) as a function of time (x-axis). Comparing survival curves of different populations allows researchers to identify which factors lead to an increased or decreased probability of survival.

Time spent:0 sec

Subject:

Biology

QID:400846

System:

Genetics and Evolution

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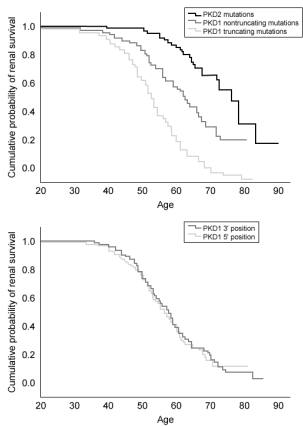


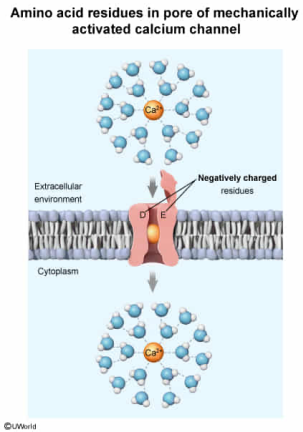
Figure 1 Probability of maintaining functional kidneys during aging, analyzed by mutation type and position

The side chains of which residues would most likely interact with calcium entering the pore of the PC2 channel? (Assume the amino acids are at physiological pH.)

- ✓ ☒ A. D and E [80%]
☐ B. R and K [9%]
☐ C. V and L [7%]
☐ D. S and F [2%]

Correct
80% answered correctly

Explanation:



Due to their **charged** nature, ions **cannot freely cross** the plasma membrane because the membrane contains a middle layer of **uncharged** fatty acid chains. Consequently, an ion must pass through the pore of an **ion channel** to enter the cell. Ion channels are transmembrane protein complexes that fluctuate between an open or closed conformation to facilitate ion transport. These channels exhibit selectivity as only **specific ions** may pass through the pore and down their concentration gradient (high to low).

For a particular ion to go through a channel, the inside of the channel's pore must be lined predominantly by **amino acid residues** with charges **opposite** to the **ion's charge** (ie, oppositely charged species **attract** each other whereas identically charged species **repel** each other). In addition, the ion must be briefly pulled out of its **hydration shell** to pass from the aqueous environment outside the plasma membrane to the aqueous environment inside the cell. The opposite charges within the pore stabilize the charge of the newly dehydrated ion.

Per the passage, PC2 forms a **mechanically activated Ca^{2+} channel**. For *positively* charged Ca^{2+} to pass, the inside of the pore must be lined by amino acid residues that are **negatively charged** at physiological pH. Therefore, of the options given, the negatively charged residues aspartate (D) and glutamate (E) are most likely to interact with Ca^{2+} entering the pore of this channel.

(Choice B) Arginine (R) and lysine (K) are both **positively charged**, and therefore would *repel* positively charged Ca^{2+} .

(Choice C) The carbon-hydrogen side chains of valine (V) and leucine (L) are **nonpolar**, and therefore unable to attract and interact with positively charged Ca^{2+} .

(Choice D) Serine (S) is **uncharged and polar** and could *weakly* attract Ca^{2+} via **ion-dipole** forces. However, phenylalanine (F) is a *nonpolar* amino acid, and therefore *unable* to attract positively charged Ca^{2+} .

Educational objective:

Ion channels allow charged atoms to pass between the extracellular and intracellular environments down their concentration gradient. The charge of the amino acids that line the inside of the channel's pore plays a key role in determining which ion can pass through.

Time spent: 0 sec
Subject:
Biology

QID: 400847
System:
Genetics and Evolution

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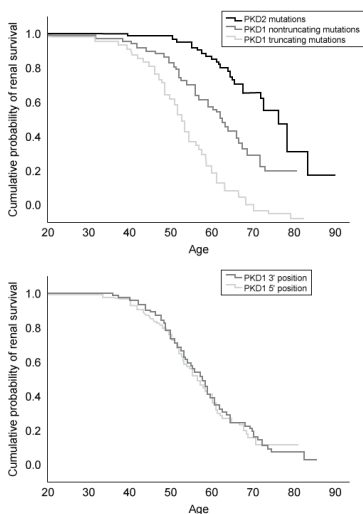


Figure 1 Probability of maintaining functional kidneys during aging, analyzed by mutation type and position

Kidney failure caused by advanced ADPKD would directly disrupt all the following components of physiological homeostasis EXCEPT:

- ☐ A. blood nitrogen levels. [7%]
- ☐ B. blood pH. [4%]
- ☒ C. leukocyte production. [72%]
- ☐ D. erythropoietin production. [15%]

Correct
72% answered correctly

Explanation:

Kidney function

- Control of extracellular fluid volume and regulation of blood pressure via the renin-angiotensin system
- Regulation of osmolarity (ie, excretion or retention of water)
- Regulation of ion concentration (Na^+ , K^+ , Ca^{2+}) by balancing dietary ion intake with urinary excretion
- Regulation of pH via bicarbonate buffer system (excretion/retention of HCO_3^- and H^+)
- Excretion of waste (creatinine, ammonia, urea, foreign substances)
- Production of renin (enzyme involved in blood pressure regulation) and hormones (erythropoietin for erythrocyte production and calcitriol for increased Ca^{2+} levels)

The kidneys' primary function is to maintain physiological homeostasis by regulating the water and ion content of the blood; this level is maintained by balancing the intake of these compounds with their excretion in urine. According to the passage, patients with advanced ADPKD may progress to kidney failure. As a result, these patients will have severely impaired kidney function and direct disruption of the following:

- **Blood nitrogen levels:** The body normally produces metabolic waste products such as creatinine from muscle metabolism, and nitrogenous urea along with uric acid. The kidneys remove both metabolic waste and foreign substances (eg, drugs, environmental toxins) by collecting these in the filtrate and excreting them in the urine. If the kidneys are unable to excrete waste, the resulting waste accumulation will become toxic and cause serious medical complications (eg, increased blood nitrogen levels due to lack of urea excretion) (**Choice A**).
- **Blood pH:** When extracellular fluid (eg, blood plasma, interstitial fluid) becomes excessively **alkaline** (high **pH**), the kidneys excrete HCO_3^- and retain H^+ , causing blood pH to decrease. Conversely, when the extracellular fluid becomes excessively acidic (low pH), the kidneys excrete H^+ and retain HCO_3^- , causing blood pH to rise. Via this process, the kidneys are able to maintain blood pH within the required narrow physiological range, which would be disrupted in ESRD (**Choice B**).
- **Erythropoietin production:** The adult kidneys normally produce erythropoietin, a hormone that signals bone marrow to increase red blood cell (erythrocyte) **production**. Generation of erythropoietin would likely be impaired in ESRD (**Choice D**).

Leukocytes (white blood cells) are **immune system cells** that serve to protect the body against infectious agents and foreign antigens. Although infections (eg, UTI) may lead to increased activity of leukocytes, production of leukocytes occurs in the **bone marrow**, *not* the kidneys. Consequently, leukocyte production would likely *not* be *directly* affected by kidney failure.

Educational objective:

The kidneys' primary function is to maintain the ion and water balance of the blood. They also play a key role in regulating multiple aspects of physiological homeostasis (eg, blood pressure, waste removal, osmolarity, blood pH, erythrocyte production).

Time spent: 0 sec
Subject:
Biology

QID: 400848
System:
Genetics and Evolution

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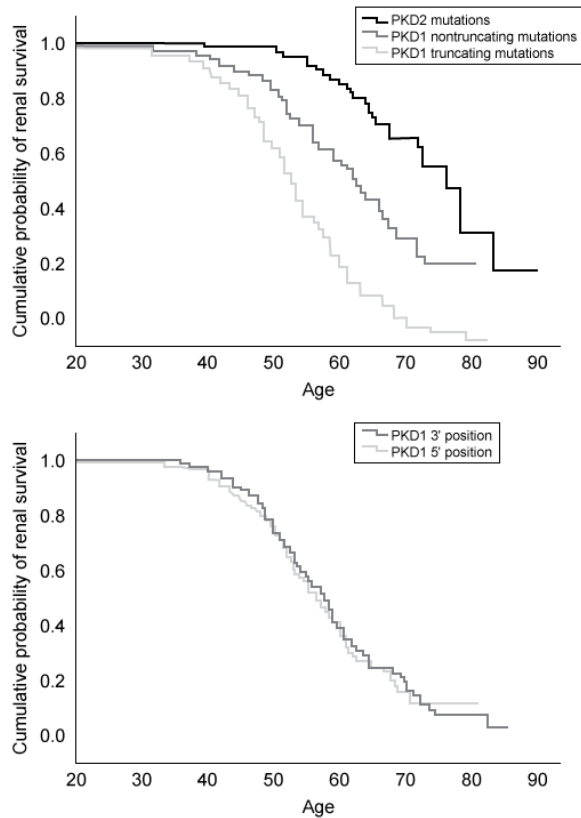


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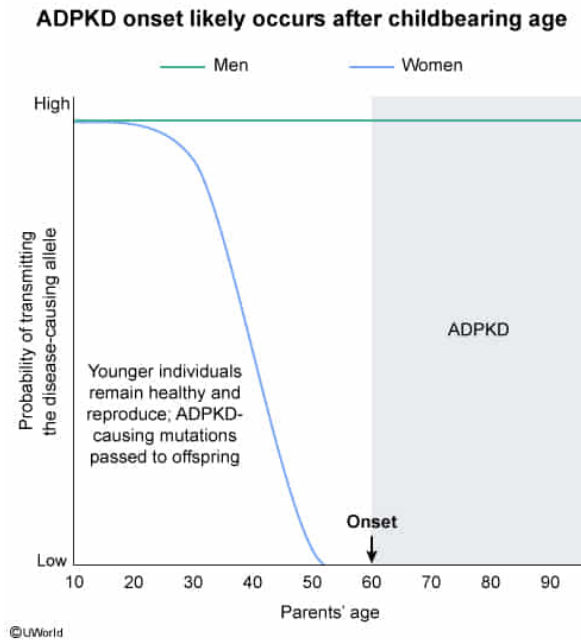
Based on the passage, is the mutated *PKD1* allele likely to be eradicated from the gene pool by natural selection?

- ☐ A. Yes, because the disease phenotype is often masked by heterozygous genotypes. [4%]
- ☐ B. Yes, because inclusive fitness is improved for individuals under age 50. [2%]
- ☐ C. No, because the mutated allele is preserved by directional selection. [10%]
- ☒ D. No, because the disease phenotype does not often emerge until after childbearing age. [82%]

Correct

82% answered correctly

Explanation:



Evolution by means of **natural selection** occurs via the **propagation** of **genes** encoding phenotypes that provide organisms with **improved survivability** and **reproductive capabilities** in a given environment (ie, increased **fitness**). Over time, the genes that manifest during the reproductive years as better adapted to the environment are selected for and are more likely to be passed on to the next generation. However, genes that are less suited to the environment or deleterious (ie, causing premature death or disease) are slowly removed from the gene pool because individuals who possess these unfavorable phenotypes are less likely to survive and reproduce.

According to Table 1, the mean age of ADPKD onset is ~58 for patients with *PKD1* mutations and ~80 for patients with *PKD2* mutations. This finding signifies that observable characteristics of the disease often do not emerge until **after childbearing age**. Although ADPKD-causing dominant alleles are deleterious, they are able to remain in the gene pool and **escape eradication** by **natural selection** because they can be passed to an affected individual's offspring **before** the individual's health and reproductive capabilities are affected.

(Choice A) If the relationship between the alleles is one of autosomal dominance (eg, ADPKD), then despite having a heterozygous genotype, the organism displays only the dominant phenotype. Therefore, a dominant allele *cannot*, by definition, be **masked** (ie, hidden).

(Choice B) The **inclusive fitness** of an individual is the sum of direct fitness (its own reproduction) and indirect fitness (its cooperative behavior to increase the number of offspring by close relatives). Therefore, inclusive fitness serves as a metric of an individual's total evolutionary success. Based on the passage, the *PKD1* mutation does *not* improve survival/reproduction of the affected individual or any of their family members.

(Choice C) In **directional selection**, one extreme phenotype is favored over all other phenotypes because of the additional fitness it confers. Consequently, the frequency of the related allele increases dramatically within a population. ADPKD does *not* improve fitness; therefore, it gains *no* directional advantage over the wild-type phenotype (ie, healthy kidneys).

Educational objective:

Deleterious dominant alleles can remain in the gene pool if the organism's fitness remains unaffected, even when the deleterious alleles reduce survival after the reproductive years. In addition, deleterious recessive alleles evade elimination by natural selection through phenotypic masking in heterozygotes.

Time spent:0 sec

Subject:
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